

CAPSTONE PROJECT

A Hybrid Hyperledger Fabric–IPFS Framework for Tamper Detection and Holder-Controlled Privacy in Academic Certificate Verification

Supervisor

Dr. Anisur Rahman

Associate Professor
East West University

Presenters

Mashiur Rahaman

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Md Adnan

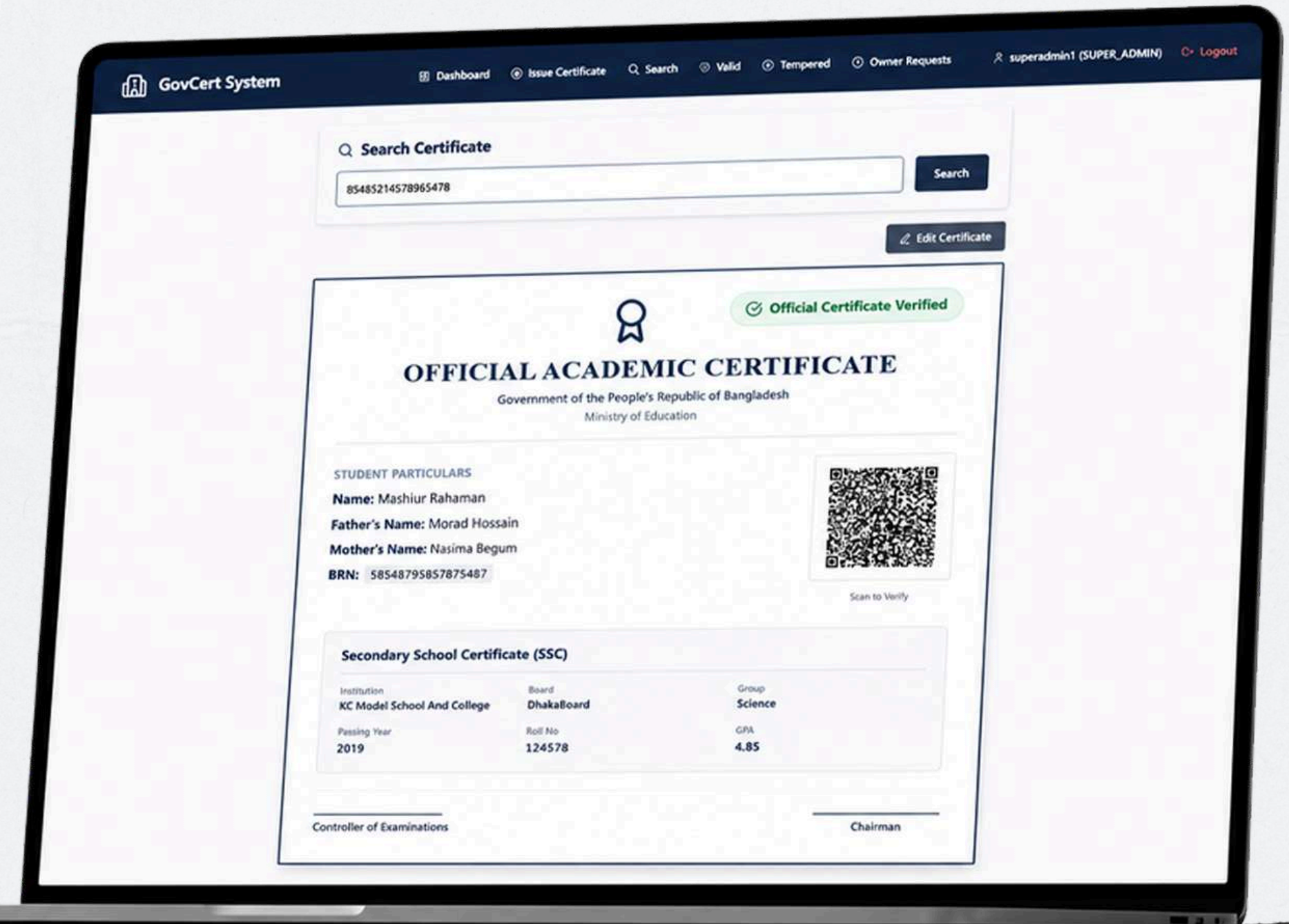
2022-3-60-179

Sheikh Sarafat Hossain

2022-3-60-109

Rijia Parveen Raya

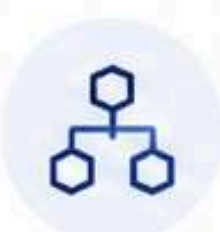
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Overview

This presentation provides an end-to-end walkthrough of our hybrid framework for secure, tamper-evident and privacy-preserving academic certificate verification using Hyperledger Fabric and IPFS.



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|-----------|--|-----------|--|
| 01 |  Problem & Motivation
Understanding the challenges in current academic certificate verification. | 05 |  Security & Privacy
Tamper detection, multi-signature governance and holder-controlled privacy. |
| 02 |  Research Gap
Limitations of existing solutions and the need for a better approach. | 06 |  Results & Analysis
Performance evaluation and experimental results. |
| 03 |  Proposed Framework
Our hybrid Hyperledger Fabric-IPFS solution for secure verification. | 07 |  Future Scope
Current limitations and possible future enhancements. |
| 04 |  System Design & Workflow
Architecture, roles, and certificate issuance & verification processes. | 08 |  Live Demo
A working demonstration of the proposed system. |



AND FINALLY...
LIVE DEMO

A live demonstration of the implemented system will be shown **at the end**.



Introduction

Verifying a certificate still means **trusting a phone call**

An employer who receives a degree certificate has no way to confirm it independently. They email the university and wait, or they accept the paper at face value.

The affected people

- ▶ Universities
- ▶ Students
- ▶ Employers
- ▶ Examination Board



The Problems

Academic certificate **forgery** is **rising** globally especially in developing countries like Bangladesh.



Centralized systems
Prone to manipulation



Expensive verification
Delays occur



Easy access to
Counterfeit certificates

Research Gaps

Current systems **fail** to provide a secure, scalable, and trustworthy solution for certificate verification.



Ethereum

High gas fees, scalability issues,
Lack of Privacy, Network Congestion



Fully On-chain Storage

Not privacy-friendly, Very Expensive,
Not scalable, Irreversibility



Centralized Databases

Single Point of Failure, Vulnerable to Attacks,
Lack of Tamper Evidence, Trust Dependency



No Institutional Control

Unauthorized Access, No Role-
Based Governance, Risk of Malicious
Activities

Proposed Solution

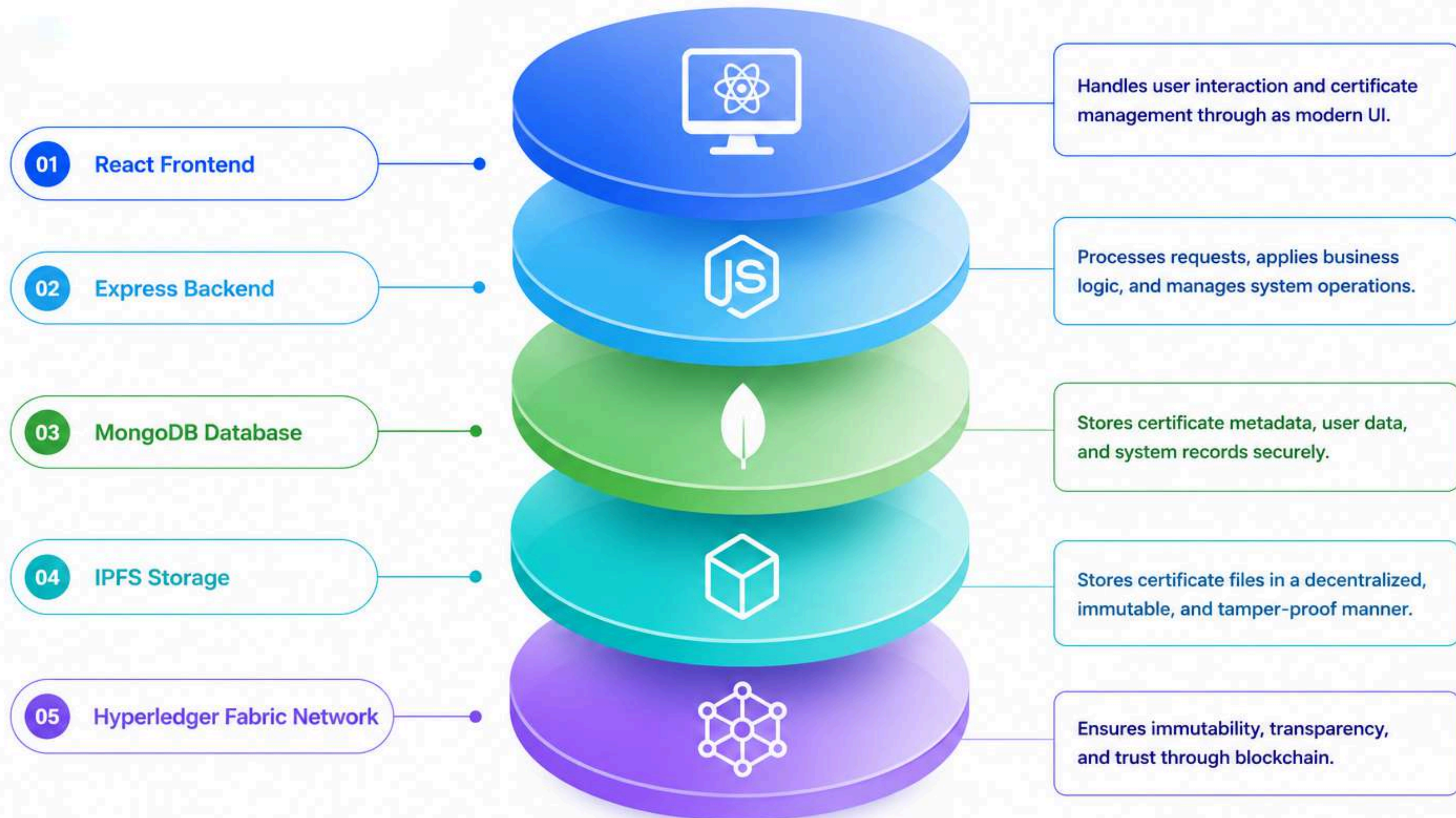
*A Hybrid Blockchain Framework for **Tamper-Evident Certificate Verification***

Core Components

- ▶ MongoDB
- ▶ IPFS
- ▶ Hyperledger Fabric
- ▶ Holder Privacy Control
- ▶ Public Verification



System Architecture



Separation of duties is enforced twice



AUTHENTICATED

Admin

Submits certificate drafts and edit requests. Cannot write to the ledger.

- ✓ Create and edit drafts
- ✓ Search and review requests
- ✓ Generate QR codes



LEDGER AUTHORITY

Super Admin

The only role whose fabric identity the chaincode accepts for state changes.

- ✓ Approve and anchor certificates
- ✓ Multi-signature governance
- ✓ Revocation and org membership



NO LOGIN

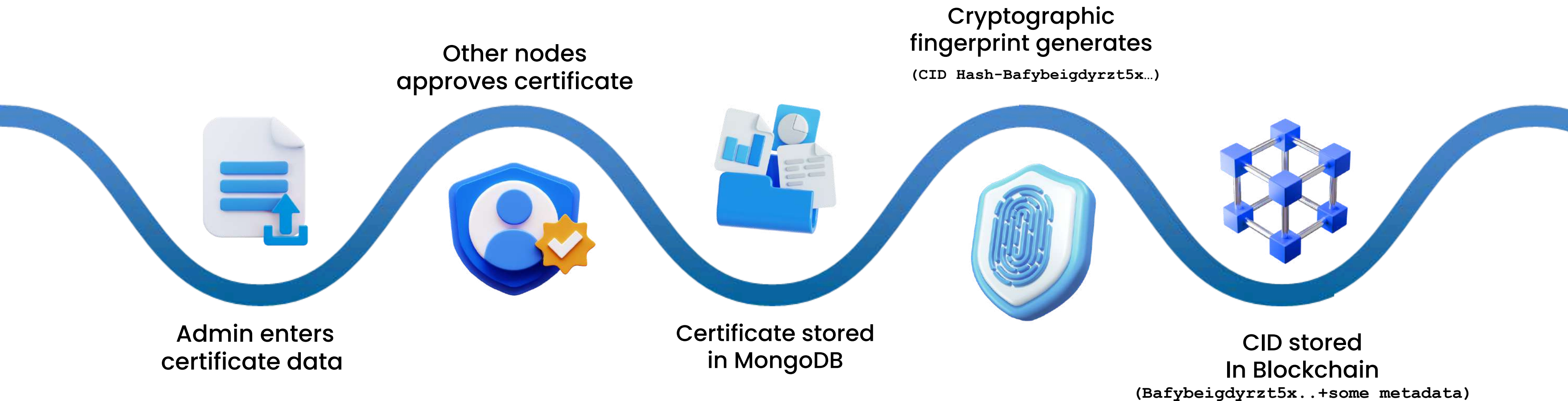
Public & holder

Verifies a certificate, or authenticates by OTP to control their own record.

- ✓ Verify by BRN or QR
- ✓ Toggle change and view locks
- ✓ Request an update

Certificate **Issuance** Process

A hybrid workflow combining validation,
content addressing, and blockchain anchoring



Certificate Issuance Process

Backend Process

Issue New Certificate

Step 1: Personal Information

What are you issuing?

SSC Certificate

HSC Certificate

BRN (Birth Registration Number) *

Full Name *

Email Address

Father's Name *

Mother's Name *

Next: Education >

Issue New Certificate

Step 1: Personal Information

What are you issuing?

SSC Certificate

HSC Certificate

BRN (Birth Registration Number) *

23456789098765432

Fetch Details

Certificate details fetched successfully.

Full Name *

Email Address

Father's Name *

Mother's Name *

Next: Education >

Issue New Certificate

Step 2: Educational Qualifications

HSC Info

Institution *

Board *

Passing Year *

Adamjee Cantonment College

DhakaBoard

2024

Roll No *

GPA *

Group *

678567

4.75

Humanities

< Back

Next: Preview >

Issue New Certificate

Step 3: Preview Information

Review Details

Name: Md Adnan

BRN: 23456789098765432

SSC Included: Yes

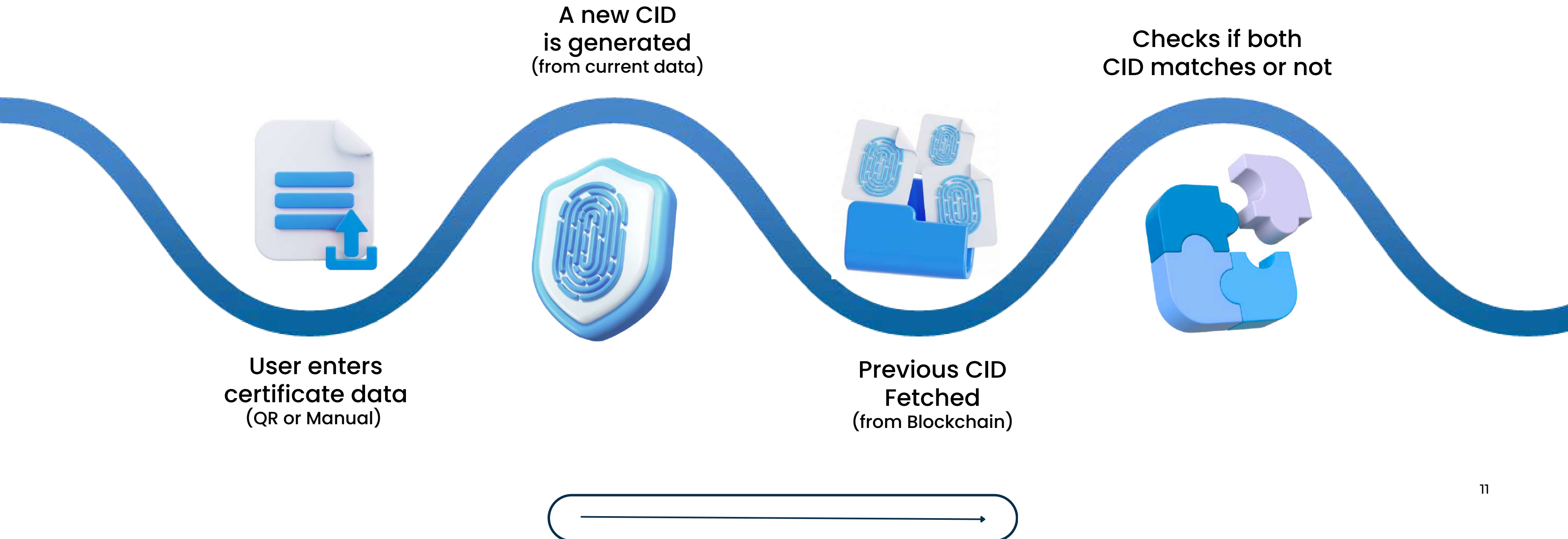
HSC Included: Yes

< Back

Issue Certificate

Certificate **Verification** Process

Ensuring certificate authenticity by
comparing generated and stored CIDs



Certificate Verification Process

Backend Process

 GovCert System

[Dashboard](#) [Tracker](#) [Search](#) [Valid](#) [Tempered](#) [Pending](#) [Multi-Sig Approvals](#) [Create Admin](#) [Revocations](#) [Network Governance](#) [Settings](#)

ahmed [Logout](#)

 Pending Approvals

All Boards 

All Exams 

All Years 

All Institutes 

3 Request(s)

BRN	NAME	TYPE	REQUESTED BY	ACTIONS
58548795857875487	Mashiur Rahaman	UPDATE	mashiursadmin	→ Compare & Approve  View
99988877766655544	Test Student	UPDATE	rajshahi_test	→ Compare & Approve  View
15426531254541256	Test dhaka dasfd	UPDATE	dhaka11	→ Compare & Approve  View

Change one character and the fingerprint stops matching

AT APPROVAL

Original record

Anchored

name: "Rijia Parveen Raya"

hscGpa: 5.00

sscGpa: 5.00

CONTENT IDENTIFIER

QmX7f2_a91c



FABRIC WORLD STATE

QmX7f2_a91c

Immutable. Cannot be edited by an administrator.

≠
Mismatch means
tampered

AFTER AN UNAUTHORIZED EDIT

Edited record

GPA changed

name: "Rijia Parveen Raya"

hscGpa: 4.75

sscGpa: 5.00

RECOMPUTED IDENTIFIER

Qm84d9_37e0

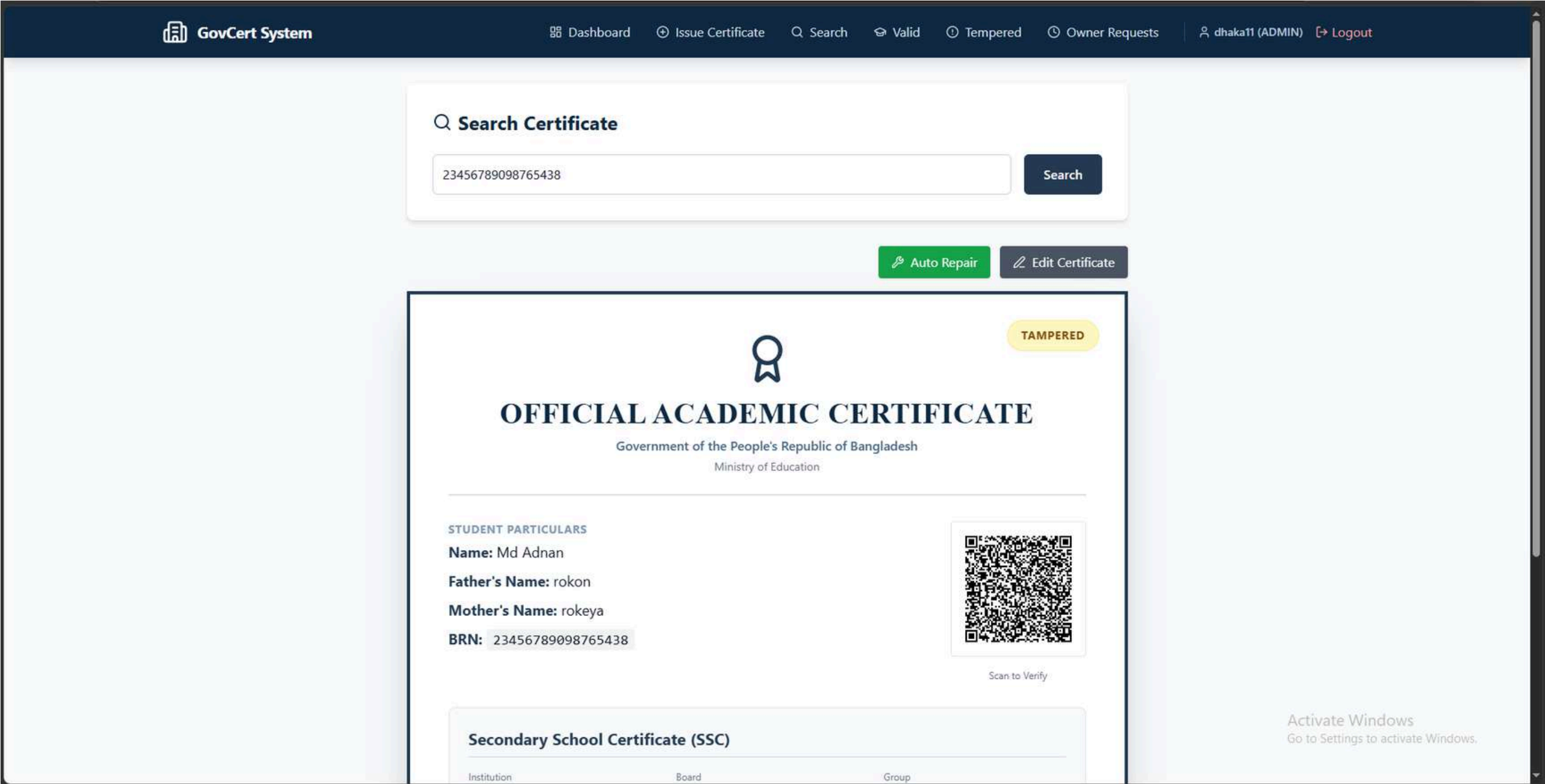


VERIFICATION RESPONSE

TAMPERED

Auto-repair can restore the original payload from IPs.

Change one character and the fingerprint stops matching



Certificate Revocation

GovCert System

DashboardTracker

produce-hartford-absolute-acquire.trycloudflare.com says

✔ Certificate Revocation Executed!

Blockchain: ✔ Revoked
Pinata/IPFS: ✔ Unpinned
MongoDB: ✔ Deleted

OK

ovals

Create Admin

Revocations

Network Governance

Settings

ahmedLogout

Revocation Approval

Multi-signature certificate revocation. Requires 2 signatures.

REVOKE: 23456789098765438

SIGNATURES: 2 / 3

Certificate Holder: Md Adnan

Requested By: sarafat

Reason: Tamper GPA

8/12/2026, 12:57:54 AM

✔ sarafat

✔ mashiursadmin

Awaiting...

⚠ Warning: Your signature will be the final approval. The certificate will be immediately revoked on the blockchain, unpinned from IPFS, and deleted from MongoDB. This action is irreversible.

Signing...

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Holder Controlled Privacy



Student



View Lock

Control who can view
your data



Change Lock

Restrict changes and
keep data safe



OTP Authentication

Verify identity with secure
one-time password



Holder Controlled Privacy

 GovCert System

Certificate Verification Portal

Government & Corporate Verification System



Verify by Details

Enter Certificate BRN and Academic Details manually to verify official records.



Verify by QR Code

Scan the QR code on the certificate to instantly verify and view the digital record.



Manage Privacy

Manage certificate change locking and public viewing separately using the holder control code.

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Manage Certificate Privacy

Certificate lock controls whether changes can be requested. View lock controls whether verified certificates are shown to others.

Certificate change lock is now UNLOCKED.

CERTIFICATE CHANGE LOCK

UNLOCKED

Changes allowed

Admins and super-admins can request certificate updates.

PUBLIC VIEW LOCK

UNLOCKED

Visible to others

Verified certificates are shown as usual during public verification.

Certificate Change Control

Unlock Certificate

Lock Certificate

Public View Control

Unlock View

Lock View

Request Certificate Update

Submit a request to change your certificate details. This request will go to the administrative queue for review.

Full Name

Leave blank to keep current

Father's Name

Leave blank to keep current

Mother's Name

Leave blank to keep current

Challenge SSC GPA

New SSC GPA

Challenge HSC GPA

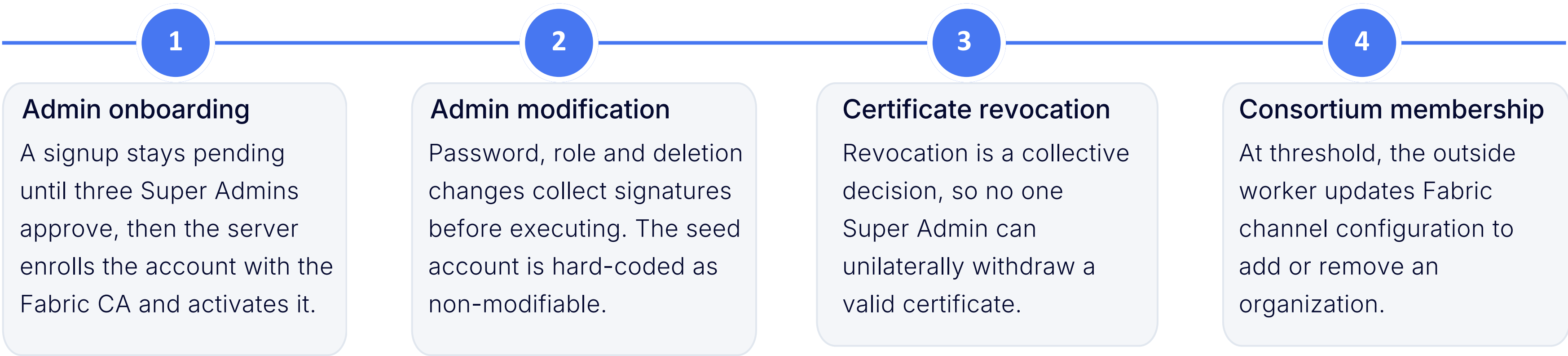
New HSC GPA

Submit Update Request

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No single administrator can **act alone** on a privileged change

Four workflows share one pattern:
an approvals array on a pending request, incremented one signature at a time, executed at threshold.



A documented divergence from the paper

The research paper specifies a five-of-N scheme. The evaluated deployment sets all four threshold constants to three, a deliberate lower-friction choice for a single-organization test network, raising them to the intended security target is listed as future work.

Result & Analysis

End-to-end issuance

4080.81 ms



MongoDB write, CID generation,
Pinata pin, and the Fabric commit.

CID generation

0.4416 ms



ipfs-only hashed over
the canonical payload.

System	No Gas Fees	Holder Privacy	IPFS Hybrid	RBAC Chaincode	Revocation	No Crypto Wallet
Ethereum-based	✗	✗	✗	✗	✗	✗
Verifi-Chain	✓	✗	✓	✗	✗	✗
Ghani et al. (Fabric)	✓	✗	✗	✓	✗	✓
Saleh et al. (Privacy Fabric)	✓	✗	✗	✓	✗	✓
Our System ★	✓	✓	✓	✓	✓	✓

Future Scope



Government Collaboration

Collaborate with relevant government authorities to validate, adopt, and deploy the system at a national scale.

01



Large-Scale Deployment

Expand the platform to support large-scale certificate issuance and verification across educational institutions.

02



Mobile Application

Develop and publish dedicated Android and iOS applications on the Google Play Store and Apple App Store, enabling convenient access for end users.

03

**THANK
YOU**

